Design Process Report – Evaluation

Objective and validation metrics

The test aimed to evaluate the immersive quality, engagement, and usability of the prototype for a virtual retail shopping application. Success was defined by positive participant engagement, perceived immersion, and constructive feedback regarding features and user experience.

Results

Feedback from Ben:

My IP was not immersive enough, as the interface was 2D despite having 2 interactions. After explaining my idea of the application, Ben suggested some feedback. He explained how this application of retail shopping can be fun and engaging, rather than following the same old traditional method. This idea begins with how a user typically scrolls on social media, such as Instagram, and then, through algorithms, you receive posts about retail shopping brands in your feed. These posts can be highly intuitive and playful. It does not necessarily have to be a proper shopping experience for the user, but more importantly, it can be immersive and engaging.

Feedback from the students:

To gain proper insights about the expectations of the virtual reality model for a retail store, I followed the technique of asking students to doodle some sketches. With tons of variety to choose from clothing, it can often lead to the consumption of a lot of time. One student came up with the idea of adding a search bar to promote fast browsing, where people can simply type or speak up, and with voice control, the searched item will pop up. Thus, making the shopping experience efficient.

The capsule in the scene would be a player where the colour already applies. However, a change in the colour of the capsule would be reflected in the mannequins to have a real effect. Also, there would be an actual 3D cart pop-up flashing, where all the items selected would be dumped into it.

Also, the virtual environment would be like walking inside a store or a closet where clothes would be hung on the rack, and accessories would be placed on the platform, and you, as an avatar, would be walking through the store.

🧥 👕 👖 emojis or actual 3d models of clothing types will pop for a realistic approach.

Analysis/Insights

From the feedback, key themes emerged. The current prototype lacks immersion due to its 2D interface, highlighting the need for a stronger 3D environment. Students emphasized the importance of efficiency in browsing, suggesting a search bar and voice control. They also proposed immersive elements such as 3D carts, interactive avatars, realistic environments, and visual cues like clothing emojis or 3D models.

Evaluation of Aims

The objective of testing immersive engagement was only partially met. While participants found the idea promising, they noted that the 2D interaction did not align with expectations for a VR-based retail store. The aims of usability and efficiency were validated through suggestions like search and voice control, but immersion was identified as lacking.

Concept Iteration

Based on the insights, the next iteration will:

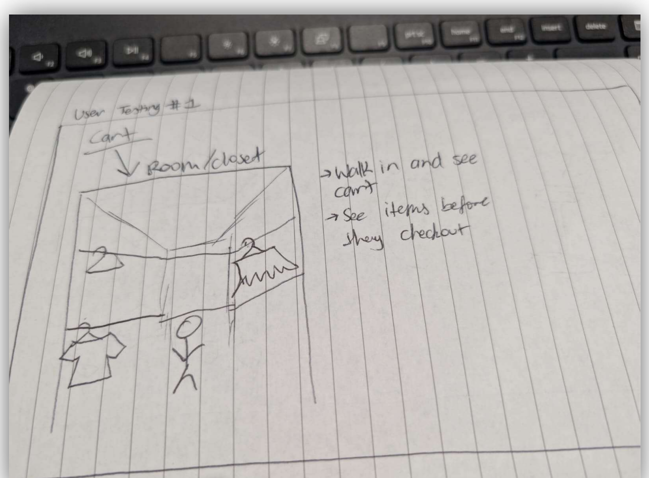
- Transition from a 2D to a 3D immersive shopping environment.
- Integrate a search bar with voice control for efficiency.
- Enhance realism by implementing avatars, 3D carts, mannequins reflecting choices, and detailed store-like environments.
- Use 3D models and emojis for better visual engagement.

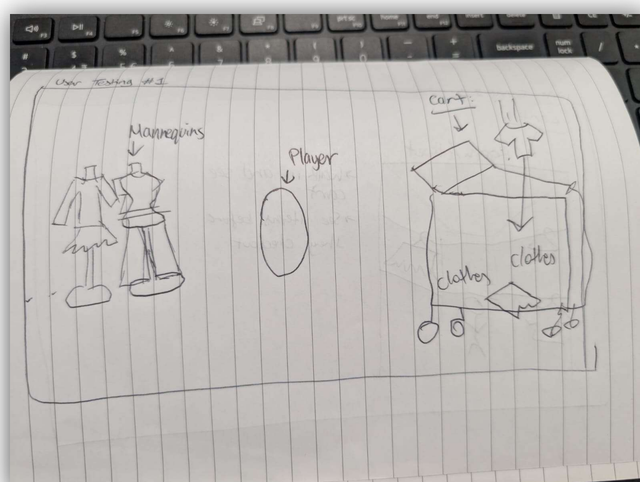
Reflection

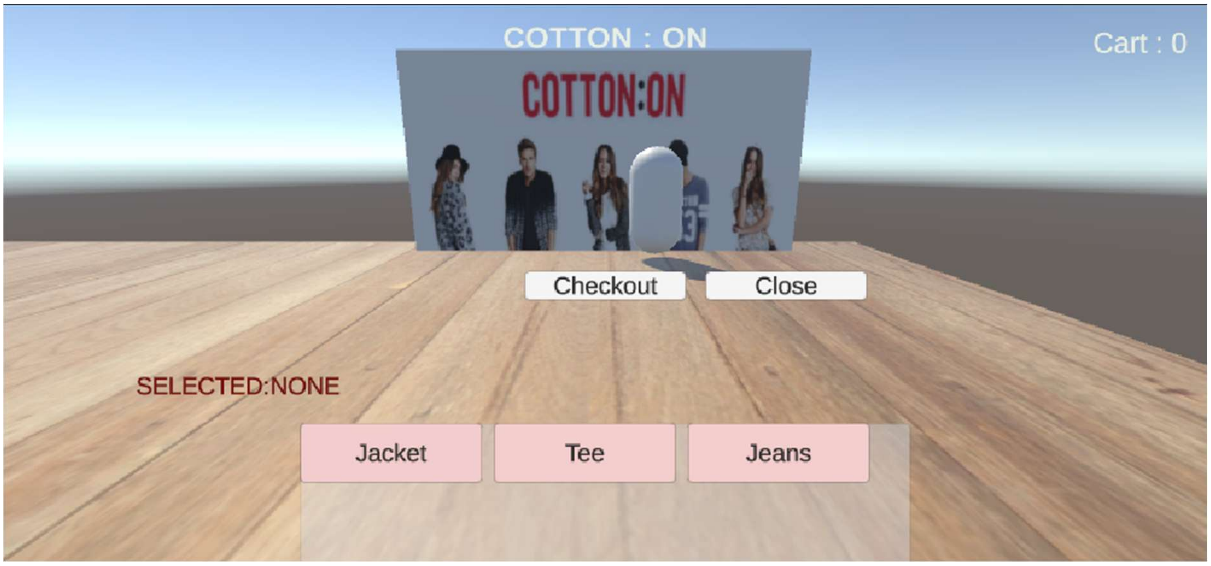
This evaluation highlighted the gap between user expectations of VR experiences and the prototype's current delivery. The process reinforced the importance of immersive design in XR applications. Future testing will focus on measuring the effectiveness of 3D interactions, user navigation, and satisfaction with immersive features.

Appendices

Below is the feedback obtained from the students post user testing.







Above is the IP 1 testing scene for the Cotton On retail shopping experience.